

CAX2 is required for the *Arabidopsis* transcriptional response to spaceflight, uncoupling gene expression from root-growth remodelling

Richard Barker¹

¹[Affiliation — TO CONFIRM], .

Contributing authors: admin@coseccloud.com;

Abstract

Spaceflight reprogrammes plant gene expression and growth, but the signalling components that translate the orbital environment into altered development remain incompletely defined. The Advanced Plant Experiment-05 (APEX-05) flew *Arabidopsis thaliana* — wild-type Col-0, a mutant in the calcium/H⁺ exchanger *CAX2* (*cax2-2*), and a mutant in the NADPH oxidase *RBOHD* (*rbohD*) — on the International Space Station alongside matched ground controls, profiling the root and shoot transcriptome (RNA-seq) and root system architecture (RSML morphometrics) from the same plate wells. Differential expression (DESeq2, flight-referenced) showed that wild-type mounts a substantial flight response (root 284, shoot 168 differentially expressed genes, DEGs), corroborated by an independent L1-regularised (LASSO) classifier (100% leave-one-out accuracy; DEG-overlap $p \leq 10^{-127}$). Projecting the bulk signal onto a published *A. thaliana* single-cell reference localised the wild-type and *rbohD* root response to the outer root — epidermis, cortex and the gravity-sensing **columella** — and implicated glutathione metabolism and phenylpropanoid biosynthesis; the response most resembled hypoxia, oxidative and defence stress programmes. Strikingly, ***cax2-2* almost entirely abolished the transcriptional flight response** (root 5, shoot 3 DEGs; $\sim 5\times$ smaller flight axis than wild-type), identifying *CAX2* as required for normal spaceflight transcriptional reprogramming. Yet spaceflight still shortened *cax2-2* roots comparably to wild-type, **uncoupling the transcriptional and morphological flight responses**. We release APEX-05 as a FAIR, reproducible data-and-code package integrating transcriptomics, root imaging and a machine-learning analysis layer.

Keywords: spaceflight, *Arabidopsis*, *CAX2*, calcium signalling, *RBOHD*, transcriptomics, root architecture, APEX-05

APEX-05 — Advanced Plant Experiment-05. **Manuscript status:** final integrated manuscript for the three-genotype analysis (Col-0, *cax2-2*, *rbohD*) on the corrected 64-sample count matrix; all quantitative results are reproducible from the repository’s **analysis/** and **results/**. [TO CONFIRM] marks fields requiring author input (metadata, hardware, citations).

1 Introduction

Plants are strong candidates for the bioregenerative life-support systems that will make long-duration and deep-space human exploration feasible, contributing food, oxygen, water recycling and psychological benefit. Realising that role requires understanding how plants develop away from Earth, yet the spaceflight environment perturbs several of the processes most central to plant growth at once — gravitropism and the gravity-sensing apparatus [1], hydrotropism and water relations, cellular redox balance, and cell-wall remodelling. Two decades of orbital experiments, consolidated in the NASA GeneLab / Open Science Data Repository (OSDR) corpus, have established that spaceflight drives broad, organ-specific transcriptional reprogramming in *Arabidopsis* and other species [2, 3]. What remains far less clear is *which* molecular events are causally required to translate the orbital environment into altered gene expression and growth: most studies catalogue what changes in a given experiment, but few test what is necessary for the response to occur at all.

Two signalling systems are strong a priori candidates for such early relays. Cytosolic calcium is among the fastest and most general second messengers in plant environmental and mechanical sensing, and its homeostasis depends on vacuolar sequestration by $\text{Ca}^{2+}/\text{H}^{+}$ exchangers; CAX2 is one such tonoplast antiporter, characterised biochemically as a low-affinity Ca^{2+} (and broader metal)/ H^{+} exchanger with a distinct pH-dependent regulation from its paralogue CAX1 [4, 5]. In parallel, apoplastic reactive-oxygen production by the NADPH oxidase RBOHD is a hub for stress and defence signalling and mediates rapid, systemic responses to diverse stimuli [6, 7]. Both calcium signalling and RBOHD-derived ROS have been proposed as early components of the plant response to microgravity, but their contributions have not been tested directly in flight by loss-of-function genetics.

The Advanced Plant Experiment-05 (APEX-05) was designed to make that test. We flew wild-type *Arabidopsis thaliana* (Col-0) alongside null mutants in *CAX2* (*cax2-2*) and *RBOHD* (*rbohD*) [TO CONFIRM: **allele line identifiers**] on the International Space Station with matched ground controls, and profiled, from the same plate wells, both the root and shoot transcriptome (RNA-seq) and root system architecture (RSML morphometrics). Integrating the two modalities through a well-level, position-paired join and a machine-learning analysis layer, we ask: (i) does removing CAX2 or RBOHD alter the magnitude or structure of the spaceflight transcriptional response; (ii) where — in which organs and cell types — is that response localised; and (iii) does the transcriptional response track the morphological one at the resolution of individual wells? We release the full analysis as a FAIR, reproducible data-and-code package.

2 Results

2.1 A corrected, position-paired 64-sample design

RNA-seq counts (33,602 genes $\times$ 64 libraries) span four genotypes $\times$ four plate wells $\times$ two tissues (root, shoot) $\times$ two conditions (spaceflight FL, ground control GC). Each well contributes matched FL and GC libraries, a **position-paired design** established from the ALSDA sample metadata via the sequencing S-numbers; this well key also joins each library to its RSML root image. A global principal-component analysis was clean — PC1 (94% of variance) separated root from shoot with no outlying libraries (Fig. 1) — supporting the corrected sample assignments. Directions throughout are flight-referenced (positive = induced by spaceflight). The fourth flown genotype, the independent *CAX2* allele *cax2-3*, was excluded after a concordance analysis showed its flight-DEG set inflated $\sim 90\text{--}100\times$ relative to *cax2-2* with an outlier library — the signature of a technical artefact rather than allelic biology (Fig. S1).

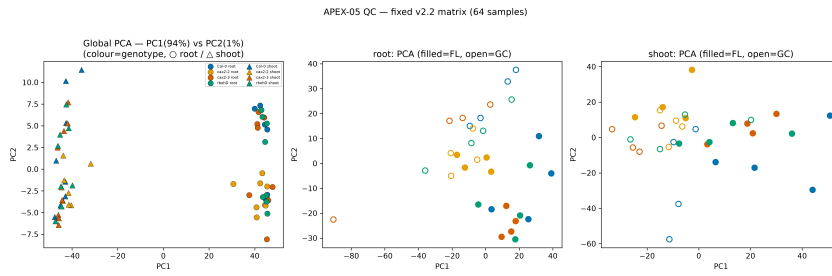


Fig. 1 Corrected 64-sample design and QC. Global PCA of log2-CPM (top 2,000 variable genes); PC1 (94%) separates root from shoot with no outliers, and within-tissue FL vs GC separation is strong in *Col-0/rbohD* and weak in *cax2-2*.

2.2 Wild-type mounts a robust flight response that CAX2 is required for

DESeq2 (FL vs GC per genotype $\times$ tissue; FDR < 0.05 , $|\log_2 \text{FC}| > 1$) detected a substantial wild-type response — **Col-0 root 284 DEGs (201 induced / 83 repressed), shoot 168 (156 / 12)** — and a comparable *rbohD* response (root 280, shoot 33; Table 1, Fig. 2). In sharp contrast, ***cax2-2* showed almost no transcriptional flight response: 5 root and 3 shoot DEGs (15 / 4 even at a relaxed 1.5-fold, FDR < 0.1 threshold)**, and its FL-vs-GC separation in morphometric-blind PCA was $\sim 5\times$ smaller than wild-type’s (Cohen-scaled centroid distance 7.9 vs 39.9 in root; Fig. 1). Because the sample QC was clean, this is a genuine biological attenuation, not a data-quality artefact: **CAX2 is required for the normal spaceflight transcriptional response.**

Wild type and *rbohD* converge on a **shared 180-gene root flight core** (of 284 and 280 DEGs respectively; Fig. S5), whereas *cax2-2* contributes essentially nothing to it (only 2 genes shared by all three genotypes). The shoot response is more

genotype-specific ($\text{Col-0} \cap \text{rbohD} = 4$ genes). An independent machine-learning check corroborated the DESeq2 calls (Fig. 3): a stability-selected LASSO logistic classifier separated FL from GC at 100% leave-one-out accuracy in every genotype, but the overlap of the LASSO signature with the DESeq2 DEGs was strong for Col-0 and *rbohD* (hypergeometric p up to 10^{-127}) and minimal for *cax2-2* (5/111 root genes), mirroring its near-absent differential expression.

Table 1 Spaceflight differential-expression (DESeq2, FL vs GC; $\text{FDR} < 0.05$, $|\log_2 \text{FC}| > 1$) DEG counts per genotype and tissue (induced / repressed).

Genotype	Tissue	DEGs	Induced	Repressed
Col-0	Root	284	201	83
Col-0	Shoot	168	156	12
<i>rbohD</i>	Root	280	198	82
<i>rbohD</i>	Shoot	33	2	31
<i>cax2-2</i>	Root	5	5	0
<i>cax2-2</i>	Shoot	3	1	2

DEG counts from `results/tables/deseq2/apex05_deseq2_DEG_counts.csv` ($\text{FDR} < 0.05$, $|\log_2 \text{FC}| > 1$; flight-referenced). The fourth flown genotype *cax2-3* is excluded (technical artefact; see text and Fig. S1).

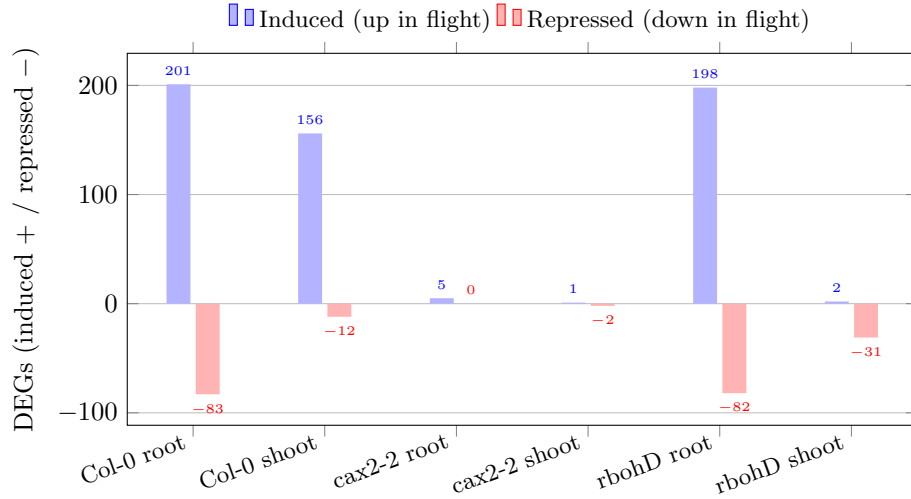


Fig. 2 Spaceflight differential expression across genotypes. DESeq2 DEG counts (induced, positive; repressed, negative) per genotype $\times$ tissue on the corrected matrix ($\text{FDR} < 0.05$, $|\log_2 \text{FC}| > 1$); Col-0 and *rbohD* respond strongly, *cax2-2* almost not at all. Rendered from `results/tables/deseq2/apex05_deseq2_DEG_counts.csv`.

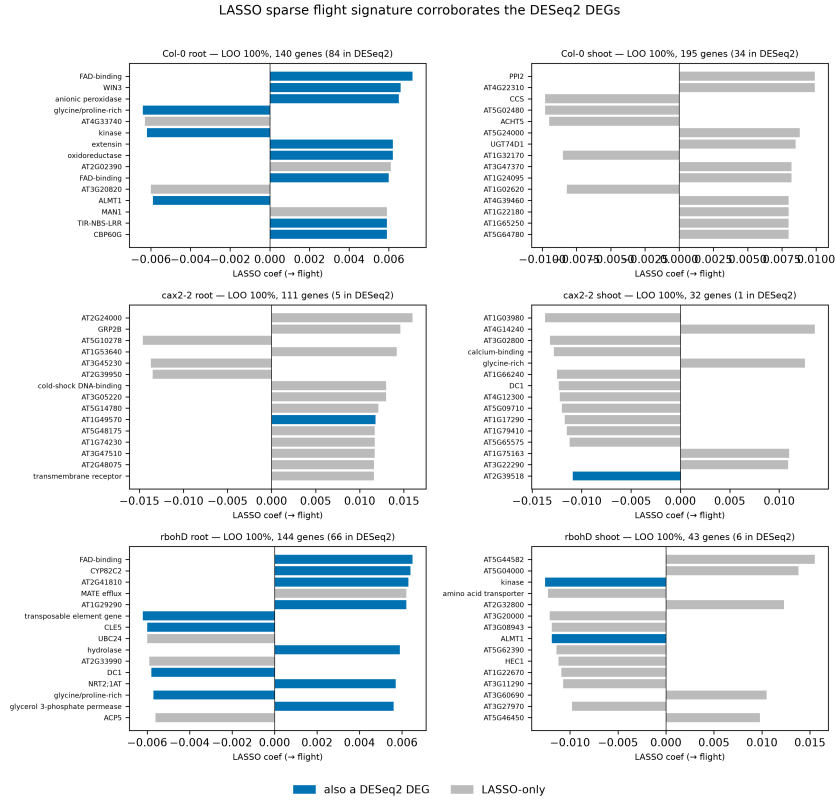


Fig. 3 LASSO corroborates the DESeq2 signal. Stability-selected sparse flight signature per genotype $\times$ tissue; overlap with DESeq2 DEGs is strong for Col-0/*rbohD*, minimal for *cax2-2*.

2.3 The wild-type flight response localises to the outer root and the gravity-sensing columella

Because APEX-05 is bulk RNA-seq, we resolved the response to cell types by projecting the DEGs onto a published *A. thaliana* single-cell marker reference (Plant Cell Marker DataBase, PCMDB; Fig. 4). The wild-type and *rbohD* root responses were significantly enriched for markers of the **non-hair epidermis, cortex and columella** (Col-0 also stele; hypergeometric $p = 8 \times 10^{-6}$ to 10^{-3}), with the *rbohD* shoot response enriched for leaf mesophyll. Painted onto root anatomy with ggPlantmap (Fig. 5), the response occupies the outer root layers in Col-0 and *rbohD* and is **blank in *cax2-2***. The enrichment of the **columella** — the root's gravity-sensing statolith tissue — is notable for a microgravity experiment.

2.4 Spaceflight remodels root architecture in all genotypes, uncoupling expression from growth

From the same wells, day-4 primary-root morphometrics showed that **spaceflight significantly shortened roots and reduced root surface and volume in**

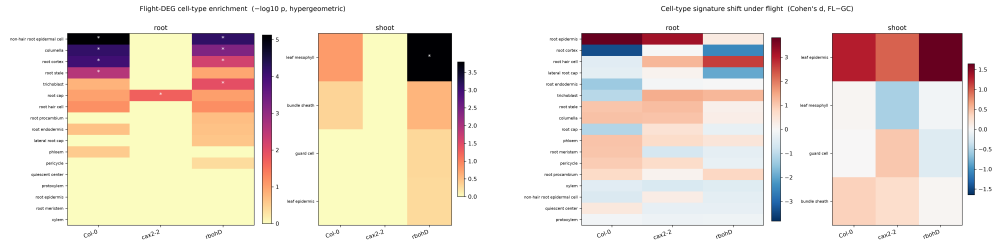


Fig. 4 Cell-type enrichment of the flight response (PCMDB projection). Hypergeometric enrichment of flight DEGs in cell-type marker sets; Col-0/*rbohD* root → epidermis, columella, cortex; *rbohD* shoot → mesophyll; *cax2-2* flat. Right: cell-type signature shift.

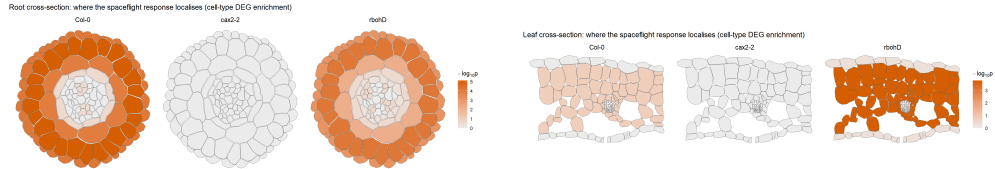


Fig. 5 Anatomical localisation (ggPlantmap). Root (left) and leaf (right) cross-section painted by cell-type flight enrichment: outer-root (epidermis+cortex) in Col-0/*rbohD*, blank in *cax2-2*.

every genotype (length/vector-length/surface/volume Cohen's $d \approx -0.3$ to -1.5 , all $p < 0.05$; *rbohD* strongest and also thinner, diameter $d = -0.69$; Fig. 6). Integrating the two modalities per well revealed a key dissociation: although *cax2-2* is the weakest responder in both assays, its transcriptome is **disproportionately attenuated relative to its retained root-shortening** (root DEGs 5 vs Col-0 284, but morphometric mean $|d|$ 0.67 vs 0.86). Thus in *cax2-2* the morphological flight response proceeds largely without the transcriptional one — **CAX2 is required chiefly for the transcriptional arm of the response**.

The position-paired design allowed this dissociation to be tested at finer resolution (Fig. 6c): across the 12 primary-genotype root wells, the per-well transcriptomic flight magnitude (mean $|\Delta \log_2 \text{CPM}|$ over the flight-DEG panel) and the per-well root-shortening ($|\Delta \text{root length}|$) were **not correlated** (Spearman $\rho = -0.30$, $p = 0.34$), and no individual flight-DEG's per-well expression change tracked root-shortening after FDR correction (0 / 385 genes at BH-FDR < 0.1). The transcriptomic axis cleanly separated *cax2-2* (low) from Col-0/*rbohD* (high), whereas root-shortening overlapped across all three — confirming, at well and gene resolution, that the space-flight root-growth response is largely **uncoupled** from the measured transcriptional response.

2.5 The flight response resembles hypoxia, oxidative and defence stress

A stress-resemblance analysis (GO “response to” over-representation, per organ; Fig. 7) found the wild-type and *rbohD* flight DEGs most resembled **hypoxia, oxidative/ROS and defence (salicylate/jasmonate/wounding)** stress programmes —

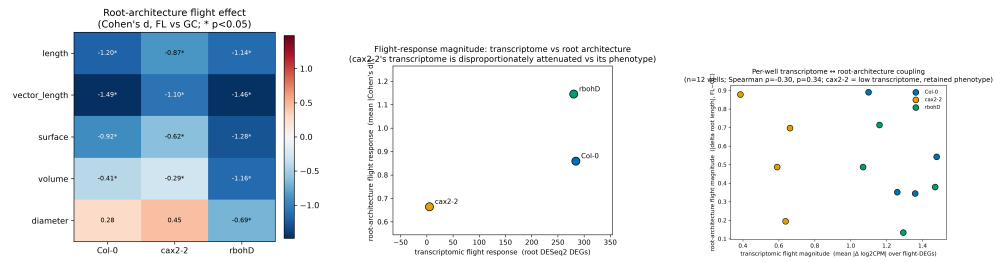


Fig. 6 Root-architecture flight response and multi-omics integration. (a) Morphometric flight effects (Cohen's d , FL vs GC); roots shorten in all genotypes. (b) Transcriptome vs phenotype magnitude (genotype level) — *cax2-2*'s transcriptome is disproportionately attenuated relative to its retained root-shortening. (c) Per-well coupling ($n = 12$ root wells): per-well transcriptomic flight magnitude vs $|\Delta$ root length| are uncorrelated (Spearman $\rho = -0.30$, $p = 0.34$).

all documented features of plant spaceflight — while *cax2-2* matched none. The resemblance was **organ-specific**: the **root** response decoded predominantly to **hypoxia and oxidative/ROS**, whereas the **shoot** response decoded to **defence/immune** programmes (biotic, salicylate in Col-0; jasmonate and wounding in *rbohD*). A cell-type-resolved decoding (Fig. S3) localised the defence/hormone signature chiefly to the **leaf mesophyll**. Systems-level KEGG analysis (Fig. 8) showed Col-0 and *rbohD* root responses converging on **glutathione metabolism and phenylpropanoid biosynthesis**; the phenylpropanoid/lignin pathway was broadly flight-induced in the Col-0 shoot. *rbohD*'s prominent glutathione-transferase and oxidoreductase signature is consistent with its role in redox homeostasis.

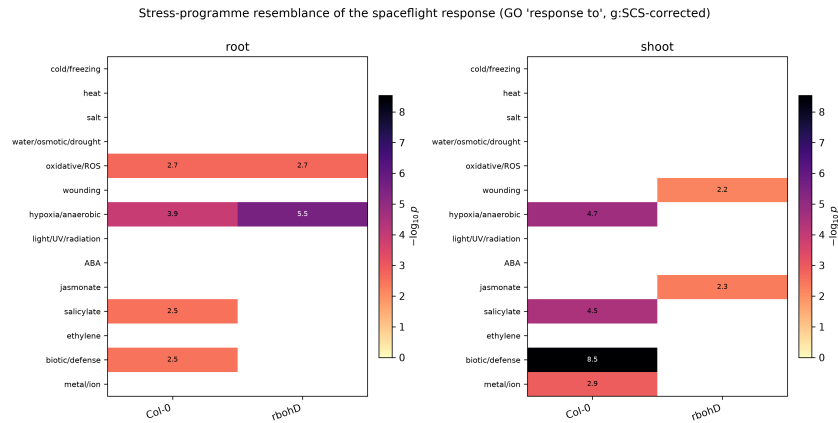


Fig. 7 Stress-programme resemblance (per organ). GO “response to” over-representation grouped into stress axes, computed separately for root and shoot: the root flight response resembles hypoxia and oxidative/ROS, the shoot response resembles defence/salicylate/jasmonate/wounding; *cax2-2* matches none.

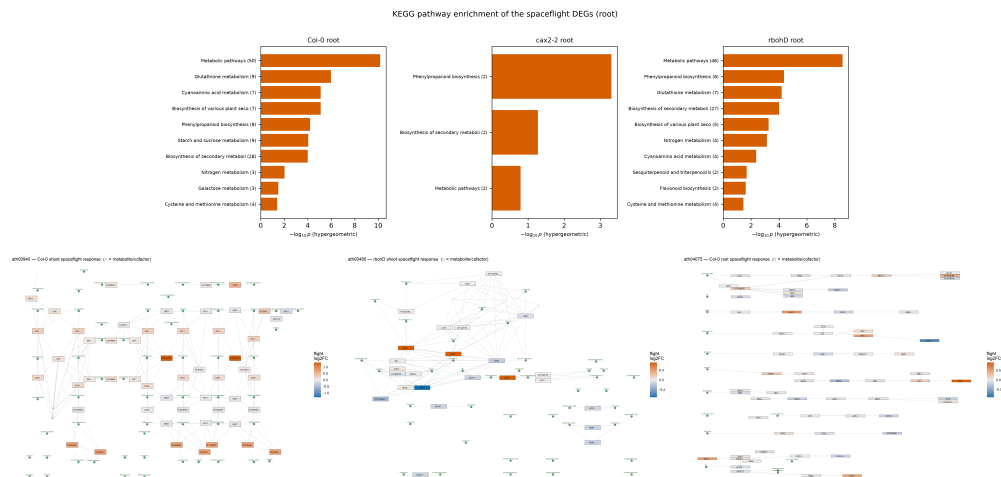


Fig. 8 Systems biology. (a) KEGG pathway enrichment (glutathione, phenylpropanoid, secondary metabolism). (b) ggKEGG pathway maps with enzyme nodes coloured by flight $\log_2\text{FC}$ and metabolite/cofactor nodes named in context: phenylpropanoid biosynthesis (Col-0 shoot, ath00940), glutathione metabolism (*rbohD* shoot, ath00480), plant hormone signal transduction (Col-0 root, ath04075).

2.6 Exploratory ligand–receptor analysis

Because cell–cell communication inference requires single-cell resolution (absent here), we tested only whether canonical *Arabidopsis* peptide ligand–receptor modules are flight-responsive in bulk (Fig. S2; explicitly exploratory). No module had both partners pass the DE threshold. Some defence-associated peptide ligands (*PROPEP3*, *PIP2*, *IDA*) were notably flight-responsive in the *rbohD* shoot, fitting the defence-stress resemblance, but true cell–cell communication mapping will require single-cell data.

3 Discussion

CAX2, but not RBOHD, is required for the transcriptional spaceflight response. The central result of APEX-05 is a genetic dissociation between two candidate signalling nodes. Wild-type Col-0 mounted a substantial, reproducible flight response (284 root and 168 shoot DEGs), corroborated by an independent LASSO classifier, and this response was largely intact in *rbohD* (280 root DEGs; a shared 180-gene root flight core with wild type). In sharp contrast, *cax2-2* was almost transcriptionally silent under flight (5 root and 3 shoot DEGs), with a flight axis roughly five-fold smaller than wild type in morphometric-blind PCA. Because sample quality was uniformly high, this is a genuine biological attenuation rather than a technical artefact. The asymmetry is informative: RBOHD-derived ROS is largely dispensable for establishing the transcriptional programme, whereas vacuolar $\text{Ca}^{2+}/\text{H}^{+}$ exchange through CAX2 is required for it — placing calcium homeostasis upstream of the bulk of

the spaceflight transcriptional response and consistent with a model in which calcium signalling is an early relay of the microgravity signal [4, 5].

The transcriptional and morphological responses are uncoupled. Despite its near-silent transcriptome, *cax2-2* still shortened its roots under spaceflight to an extent comparable with wild type, and across the 12 position-paired primary-genotype root wells the per-well transcriptomic flight magnitude and per-well root-shortening were uncorrelated (Spearman $\rho = -0.30$, $p = 0.34$), with no individual flight-DEG tracking root-shortening after correction. Thus the spaceflight root-growth response can proceed largely without the measured transcriptional one. This dissociation implies a substantial transcription-independent component to root-architecture remodelling — whether pre-existing, post-transcriptional, or mediated by rapid ionic and mechanical signalling that does not require new transcription — and cautions against inferring morphological phenotypes directly from DEG counts.

The response is anchored in gravity-sensing and environmental-interface tissues. Projecting the bulk signal onto a single-cell marker reference localised the wild-type and *rbohD* root response to the outer root — non-hair epidermis, cortex and, notably, the gravity-sensing columella — with the shoot response enriched in leaf mesophyll. That a microgravity experiment should converge on the columella, the statolith-bearing tissue of graviperception, is striking and consistent with the loss of a unit-gravity reference being a primary cue. Functionally, the flight response most resembled hypoxia, oxidative/ROS and defence stress programmes and converged at the pathway level on glutathione metabolism and phenylpropanoid biosynthesis — programmes repeatedly reported in plant spaceflight [3] — with an organ split between a root hypoxia/oxidative signature and a shoot defence/hormone signature. The prominent glutathione-transferase and oxidoreductase signature in *rbohD* is consistent with its role in redox homeostasis, and may reflect compensation for altered ROS production.

Limitations. Several caveats bound these conclusions. The transcriptome is bulk RNA-seq, so cell-type localisation rests on marker projection rather than single-cell measurement; the ligand-receptor analysis is explicitly exploratory and under-powered in bulk tissue; morphometrics are day-4 primary-root traits from a single timepoint; and the datasets, while position-paired, remain modest in replication. The mechanistic ordering we infer is a genetic-requirement argument, not a direct measurement of calcium or ROS dynamics in flight.

Outlook. APEX-05 identifies CAX2-dependent calcium handling as a required node for the *Arabidopsis* transcriptional response to spaceflight while showing that root-growth remodelling is at least partly transcription-independent. Testing this model will require complementary proteomic and metabolomic measurements, time-resolved and single-cell transcriptomics, and direct calcium/ROS reporters in flight or in ground-based analogues, together with an assessment of whether modulating calcium signalling can tune plant performance in space.

4 Methods

- **Plant material & spaceflight.** Col-0, *cax2-2* and *rbohD* [TO CONFIRM: allele/line identifiers, seed source]; ISS spaceflight vs ground control [TO CONFIRM: hardware, media, light, temperature, flight profile, fixation].
- **Design & metadata.** 4 genotypes $\times$ 4 wells $\times$ 2 tissues $\times$ 2 conditions (64 libraries). The rep $\rightarrow$ well mapping and the RNA-seq $\leftrightarrow$ RSML bridge are resolved from the ALSDA sequencing S-numbers.
- **RNA-seq & differential expression.** Reads aligned with HISAT [8] [TO CONFIRM: versions/genome]; differential expression with DESeq2 (pydeseq2) [9, 10], contrast FL vs GC per genotype $\times$ tissue, flight-referenced; DEGs at FDR < 0.05 and $|\log_2 \text{FC}| > 1$.
- **LASSO.** L1-penalised logistic regression [11] on log2-CPM, most-parsimonious penalty by leave-one-out CV, stability-selected signature ($\geq 75\%$ of LOO refits); DEG overlap by hypergeometric test.
- **Cell-type projection.** PCMDB [12] (Zenodo 10.5281/zenodo.5101271) cell-type-specific markers; hypergeometric DEG enrichment + per-sample signature scoring; ggPlantmap [13] anatomy.
- **Morphometrics & integration.** RSML day-4 primary-root traits; FL-vs-GC effect sizes (Cohen’s d , Mann–Whitney); well-level join to the transcriptome.
- **Stress resemblance & pathways.** g:Profiler [14] GO “response to” axes computed per organ and (via PCMDB markers) per cell type; KEGG [15] pathway membership (KEGG REST) and ggkegg [16] maps.
- **Reproducibility.** Python + R pipeline in `analysis/`; dependencies in `analysis/ml/requirements.txt`; single random seed; outputs in `results/`.

Data and code availability. Data & code: https://github.com/dr-richard-barker/APEX05_results_and_code (FAIR; MIT-licensed). Reference resources: PCMDB (Zenodo 10.5281/zenodo.5101271); KEGG; g:Profiler; Plant Reactome (Gramene) [17]. Sequencing archive and Zenodo DOI: [TO CONFIRM].

Acknowledgements. [TO CONFIRM: acknowledgements, author contributions, competing interests, funding.]

References.

References

- [1] Morita, M. T., Nishimura, T. & Shikata, H. Gravity sensing for gravitropism. *Annual Review of Plant Biology* **77**, 425–448 (2026).
- [2] Paul, A.-L., Zupanska, A. K., Schultz, E. R. & Ferl, R. J. Organ-specific remodeling of the Arabidopsis transcriptome in response to spaceflight. *BMC Plant Biology* **13**, 112 (2013).
- [3] Barker, R. *et al.* Meta-analysis of the space flight and microgravity response of the Arabidopsis plant transcriptome. *npj Microgravity* **9**, 21 (2023).

- [4] Shigaki, T., Pittman, J. K. & Hirschi, K. D. Manganese specificity determinants in the Arabidopsis metal/H⁺ antiporter CAX2. *Journal of Biological Chemistry* **278**, 6610–6617 (2003).
- [5] Pittman, J. K., Shigaki, T. & Hirschi, K. D. Evidence of differential pH regulation of the Arabidopsis vacuolar Ca²⁺/H⁺ antiporters CAX1 and CAX2. *FEBS Letters* **579**, 2648–2656 (2005).
- [6] Torres, M. A., Dangl, J. L. & Jones, J. D. G. Arabidopsis gp91phox homologues AtrbohD and AtrbohF are required for accumulation of reactive oxygen intermediates in the plant defense response. *Proceedings of the National Academy of Sciences* **99**, 517–522 (2002).
- [7] Miller, G. *et al.* The plant NADPH oxidase RBOHD mediates rapid systemic signaling in response to diverse stimuli. *Science Signaling* **2**, ra45 (2009).
- [8] Kim, D., Langmead, B. & Salzberg, S. L. HISAT: a fast spliced aligner with low memory requirements. *Nature Methods* **12**, 357–360 (2015).
- [9] Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* **15**, 550 (2014).
- [10] Muzellec, B., Teleńczuk, M., Cabeli, V. & Andreux, M. PyDESeq2: a python package for bulk RNA-seq differential expression analysis. *Bioinformatics* **39**, btad547 (2023).
- [11] Tibshirani, R. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B* **58**, 267–288 (1996).
- [12] Jin, J. *et al.* PCMDB: a curated and comprehensive resource of plant cell markers. *Nucleic Acids Research* **50**, D1448–D1455 (2022).
- [13] Jo, L. & Kajala, K. ggPlantmap: an open-source R package for the creation of informative and quantitative ggplot maps derived from plant images. *Journal of Experimental Botany* **75**, 5366–5376 (2024).
- [14] Raudvere, U. *et al.* g:profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update). *Nucleic Acids Research* **47**, W191–W198 (2019).
- [15] Kanehisa, M. & Goto, S. KEGG: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Research* **28**, 27–30 (2000).
- [16] Sato, N. & Tsuyuzaki, K. ggkegg: analysis and visualization of KEGG data utilizing the grammar of graphics. *Bioinformatics* **39**, btad622 (2023).
- [17] Naithani, S. *et al.* Plant reactome: a knowledgebase and resource for comparative pathway analysis. *Nucleic Acids Research* **48**, D1093–D1103 (2020).

Supplementary figures

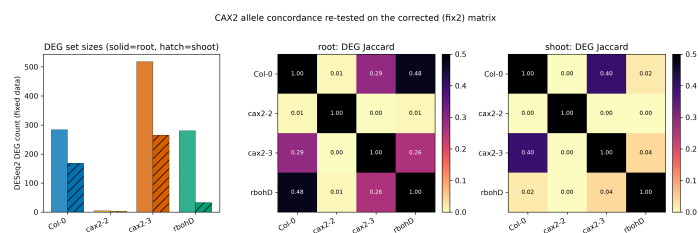


Fig. S1 *cax2-3* exclusion (corrected data). The excluded fourth genotype's flight-DEG set inflated ~ 90 – $100\times$ relative to *cax2-2* with an outlier library — a technical artefact.

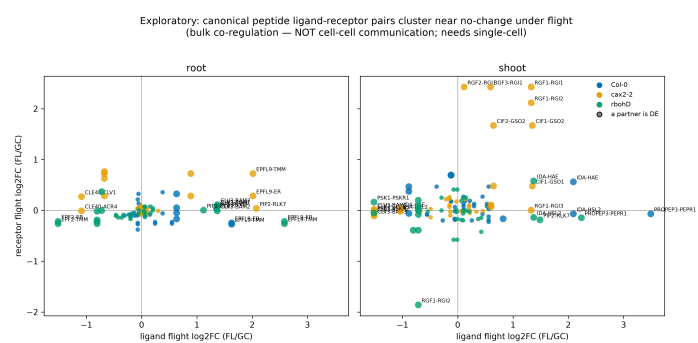


Fig. S2 Exploratory peptide ligand–receptor flight-responsiveness. No module had both partners pass the DE threshold; peptide-signalling genes cluster near no-change.

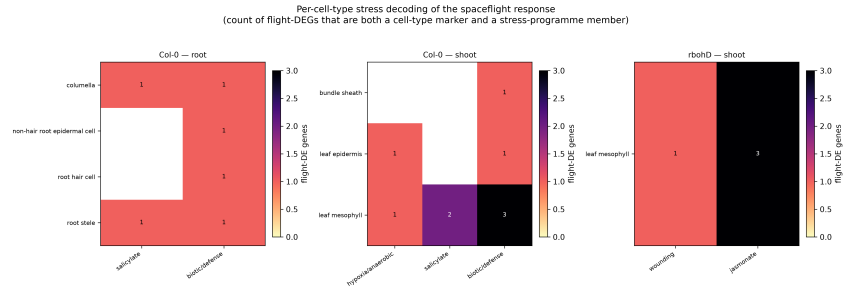


Fig. S3 Cell-type-resolved stress decoding. For each significant stress programme (Fig. 7), member flight-DEGs are attributed to cell types via the PCMDB markers. The defence/hormone signature localises chiefly to leaf mesophyll.

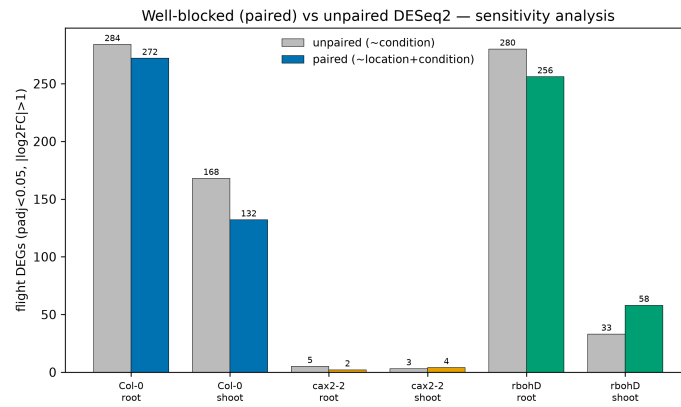


Fig. S4 Well-blocked (paired) DESeq2 — sensitivity analysis. Flight-DEG counts from the unpaired model (~condition) vs a paired model blocking on plate well (~location + condition). Counts are broadly concordant and every qualitative conclusion is unchanged.

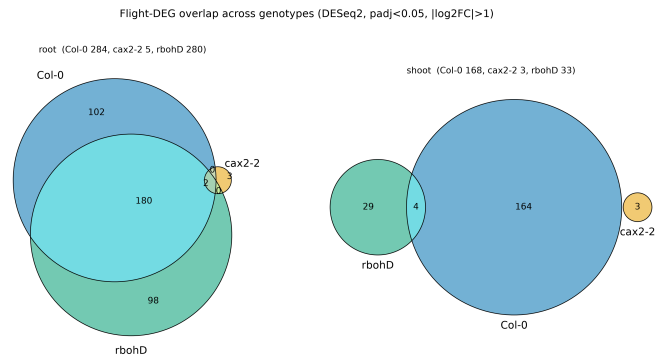


Fig. S5 Flight-DEG overlap across genotypes. Three-way Venn of the DESeq2 flight-DEG sets (Col-0, *cax2-2*, *rbohD*) for root and shoot. Wild type and *rbohD* share a 180-gene root flight core; only 2 genes are shared by all three; *cax2-2* is near-empty.